



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Computing and Applied Sciences Department Computer Science & Information Technology

Semester 4th Degree Program BS-CS Batch 2020F

Course Code: CS-224 Course Title: Theory of Automata & Formal Languages Credit Hours: 3+0

Timings: 3 Hours Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	Name	'A' Copy Serial No.

Q.No.1**8 Marks**

Read the following languages carefully and perform the given tasks.

- All words of languages whose length is completely divisible by 4 are valid.
- All words that have at least two characters length with **aa** or **bb** as a mandatory pattern.

Tasks to perform on the given strings:

- Make the Context free Grammar (CFG)
- Perform the Leftmost and rightmost derivations
- Draw the parse tree

Input Strings are:

a> aaabbbbaa
b> ababbab

Q.No.2**12 Marks**

Convert the given CFG into Chomsky Normal Form

a> $S \rightarrow ABb$ b> $S \rightarrow abcXY$ c> $S \rightarrow ABC$ $A \rightarrow aAb \mid baa$ $X \rightarrow Ybc \mid ca \mid cY \mid Y$ $A \rightarrow aaA \mid B \mid \epsilon$ $B \rightarrow bBa \mid ba$ $Y \rightarrow bY \mid \epsilon$ $B \rightarrow bB \mid C \mid \epsilon$ $C \rightarrow \epsilon$



Sir Syed University of Engineering & Technology

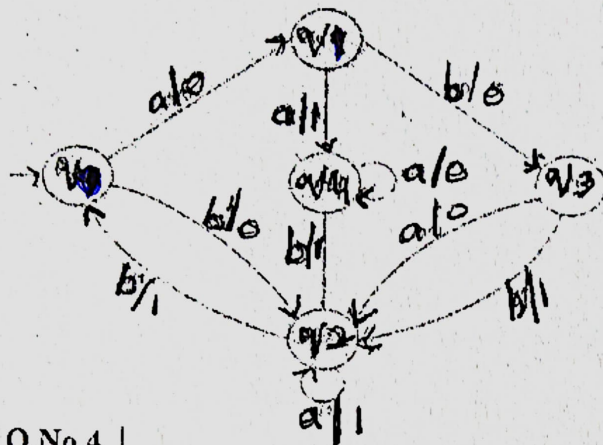
END SEMESTER EXAMINATIONS SPRING 2022

Q.No.3

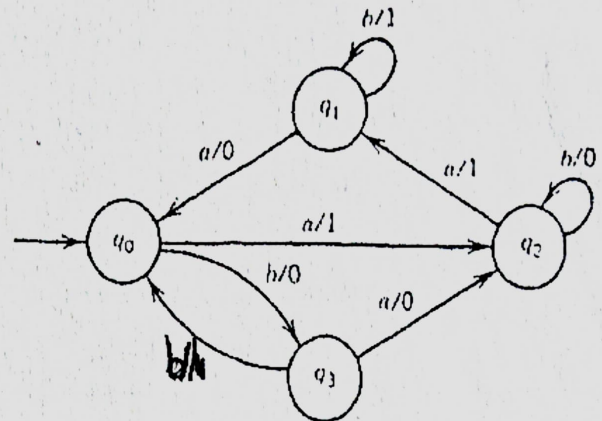
6 Marks

Convert the given Mealy machines into Moore machines

a>



b>



Q.No.4

12 Marks

Write the CFG and draw the Push Down Automata for the given languages

a> $e^x a^n b^m c^m d^y$ | $m, n \geq 0, x, y \geq 0$ b> $a^n b^{n+2}$ | $n \geq 1$ c> $a^{m+2} b^n c^n d^{p+2}$ | $n \geq 0, m, p \geq 0$

Q.No.5

6 Marks

Write the regular expression for the given CFG?

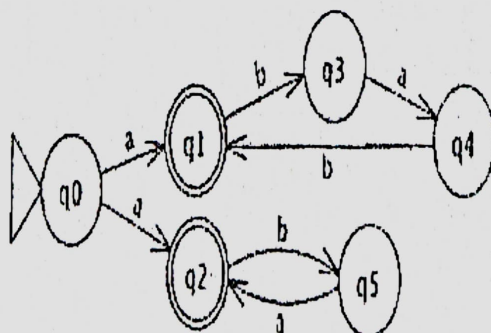
a> $S \rightarrow YXYXY$ $X \rightarrow aa \mid \epsilon$ $Y \rightarrow bY \mid \epsilon$ b> $S \rightarrow ABA$ $A \rightarrow 0A \mid \epsilon$ $B \rightarrow 11B \mid 11$

Q.No.6

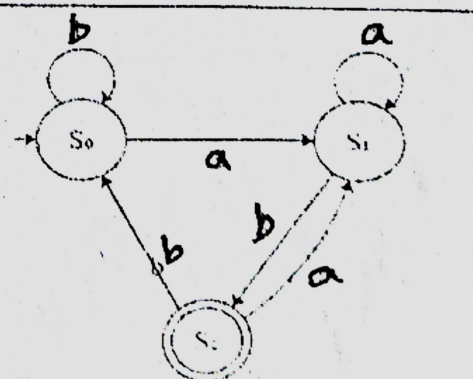
6 Marks

Convert the given DFA/NFA into Regular Expression by using transection Graph

a>



b>





Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: FoCAS Department: Computer Science

Semester 4th Degree Program BS(CS) Batch 2020F

Course Code: MS-208 Course Title: Multivariate Calculus Credit Hours: 3

Timings: 3 Hours

Date of Examination: 30-06-22

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	Name	'A' Copy Serial No.

Q. No. 1 10 Marks

a) Evaluate $\iint_S F \cdot n \, dS$ where $F = 4xzi - y^2j + yzk$ and S is the surface of the cube bounded by $(0,0,0)$ to $(1,1,1)$. (5 Marks)

b) Prove that $\nabla r^n = nr^{n-2} \mathbf{r}$. (5 Marks)

Q. No. 2 10 Marks

a) Using Fourier Series for the function $f(x) = x^2/4$ $(-\pi < x < \pi)$ show that $\frac{\pi^2}{12} = 1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \dots$ (5 Marks)

b) Transform the equation $x^2 + y^2 + z^2 = 10$ into spherical Coordinates. (5 Marks)

Q. No. 3 10 Marks

a) Find the Fourier series of the following. (5 Marks)

$$f(x) = \begin{cases} x & \text{if } -\pi/2 < x < 0 \\ \pi - x & \text{if } 0 < x < \pi/2 \end{cases}$$



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

b) Find the Fourier cosine series. (5 Marks)

$$f(x) = x^2 \quad (0 < x < p) \quad \text{Ans: } \left(\cos n \frac{\pi x}{p} \right)$$

Q. No. 4

10 Marks

a) Verify Green's theorem in the plane for $\oint (3x^2 + 8y^2)dx + (4y - 6xy)dy$ where c is the closed curve of the region bounded by $\sqrt{x}=y$ and $x^2=y$. (5 Marks)

b) Evaluate $\iint_D 2e^{x^2+y^2} dA$, D is the portion of the region between the circles $x^2 + y^2 = 25$ and $x^2 + y^2 = 49$ that lies in the first quadrant. (5 Marks)

Q. No. 5

10 Marks

a) By using stoke's theorem for the vector field $F = (2x - y)i - yz^2j - yzk$ over the upper half surface of $x^2 + y^2 = 1 - z^2$ and C is its boundary. (5 Marks)

b) Find $\int_1^3 \int_{-1}^1 \int_2^4 3(x^2y + y^2z) dz dy dx$. (5 Marks)



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Faculty of Computing & Applied Sciences Department: Computer Science & Information Technology

Semester 4th Degree Program BS(CS) Batch 2020F

Course Code: CS-222T Course Title: Introduction to Operating System Credit Hours: 3+0

Timings: 3 Hours Date of Examination: 05-07-2022 Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Attempt all questions.

Roll No.	Name	'A' Copy Serial No.
----------	------	---------------------

Q.No.1 (5+5 Marks)

- (a) What is the meaning of the term busy waiting? What other kinds of waiting are there in an operating system? Can busy waiting be avoided altogether? Explain your answer.
- (b) Describe the basic idea of the Bakery algorithm using your own words. You should mention which problem the algorithm is trying to solve, and how it is solving it (conceptually). There is no need for pseudo code.

Q.No.2 (5+5 Marks)

- (a) Given six memory partitions of 150 Kb, 300 Kb, 350 Kb, 70 Kb, 125 Kb, 200 Kb (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of 200 Kb, 300 Kb, 25 Kb, 155 Kb and 100 Kb (in order)? Which algorithm makes the most efficient use of memory?
- (b) Consider a system in which a program can be separated into two parts: code and data. The CPU knows whether it wants an instruction (instruction fetch) or data (data fetch or store). Therefore, two base-limit register pairs are provided: one for instructions and one for data. The instruction base-limit register pair is automatically read-only, so programs can be shared among different users. Discuss the advantages and disadvantages of this scheme.



Sir Syed University of Engineering & Technology END SEMESTER EXAMINATIONS SPRING 2022

Q.No.3 (5+5 Marks)

- (a) Consider the following page reference string: 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6. Assuming frame size 4, how many page faults would occur for the following page replacement algorithms? FIFO, LRU, Optimal page replacement algorithm.
- (b) Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.

Q.No.4 (5+5 Marks)

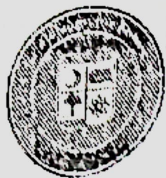
- (a) To build a robust distributed system, you must know what kinds of failures can occur.
- List three possible types of failure in a distributed system
 - Specify which of the entries in your list also are applicable to a centralized system.
- (b) A distributed system has two sites, A and B. Consider whether site A can distinguish among the following:
- B goes down.
 - The link between A and B goes down.
 - B is extremely overloaded, and its response time is 100 times longer than normal.
- What implications does your answer have for recovery in distributed systems?

Q.No.5 (5+5 Marks)

- (a) Consider a system with 5 processes P1, P2, P3, P4 and P5. Arrival time of processes is 0,1,2,3,4 with burst time requirements of these processes is 5, 3, 1, 2 and 3 respectively. Calculate average turnaround time and average waiting time. Assuming Round Robin Scheduling with Time Quantum is 2.
- (b) Consider 4 processes P1, P2, P3, P4 with arrival time 0,1,2,3 and burst time 6,4,6,1 respectively. If the CPU scheduling policy is SJF preemptive, calculate the average waiting time and average turnaround time.

4.25

8



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Ms. Sabeen Fatima Ms. Jawaria Hafeez Department CS&IT

Semester 4th Degree Program BSCS Batch 2020F

Course Code: CS-226 Course Title: Introduction to Software Engineering Credit Hours: 3+0

Timings: 3 Hours Date: 2nd July 2022 Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.

Roll No.	Name	A' Copy Serial No.

Q.No.1

(15 Marks)

Write possible test cases for the given Interface

Q.No.2

(05 Marks)

Software quality management is concerned with ensuring that software meets its required standards. What do you think "Required Standard" refers to? Explain with the help of some software applications.



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Q.No.3

(10 Marks)

As a Project Manager you can analyze given risks in your project, what will be your strategy to manage and monitor these risks?

Risk ID	Risk Description	Likelihood of the risk occurring
001	Project purpose and need is not defined	Medium
002	Project design and deliverable definition is incomplete	High
003	Project conflicts not resolved in a timely manner.	Low
004	Project schedule is not clearly understood	Low

Q.No.4

(10 Marks)

WhatsApp is an American freeware, cross-platform messaging and voice over IP service owned by Facebook, Inc. It allows users to send text messages and voice messages, make voice and video calls, and share images, documents, user locations, and other media. Suppose you have to create product with same functionalities dedicated for SSUET only. Your product will be beneficial for all the departments to share their information locally. For following a plan, Prepare Inspection Checklist makes sure to write complete checklist with detailed evaluation of process. Parameters for Inspection are as follows:

1. Data faults
2. Control Faults
3. Input/output Faults.
4. Interface Faults
5. Storage Management Faults
6. Exception Management Faults

Q.No.5

(10 Marks)

Justify the below statements with appropriate examples:

- a) Some people argue that developers should not be involved in testing their own code but all testing should be the responsibility of a separate team. Give arguments for and against testing by the developers themselves.
- b) The focus may be 'fitness for purpose' rather than specification conformance.



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Faculty: Computing and Applied Sciences Department Computer Science & Information Technology

Semester 4th Degree Program BS-CS Batch 2020F

Course Code: CS-229T Course Title: Database System Credit Hours: 3+1

Timings: 3 Hours

Max Marks: 50

Instructions:

- Please don't write anything on the Question Paper except for your Roll No. and Name.
- The paper must be solved / attempted on Printed Answer books provided by the Examinations Department.
- Please attempt all questions.

Roll No.	Name	'A' Copy Serial No.

Q.No.I.A	07 Marks
----------	----------

Normalization is the process of decomposing unsatisfactory relations by breaking up their attributes into smaller relations. Discuss your normalization choice(s) with proper justification by applying Normalization forms (1NF, 2NF) to below mentioned Sales Records in order to remove anomalies, redundancy and improve data integrity.

Sales Records

CustName	Product	ShippingAddress	NewsLetter	Supplier	SuppPhone	Price
Riaz Hussain	TV	Mall Road Lahore	Sony News	Sony	04245678	50000
Rashid Khan	Refrigerator	Electronic Market Karachi	Dawlance News	Dawlance	033324556	45000
Saleem Ahmed	TV, Audio Device	Mall Road Lahore	Sony News	Sony	04245678	50000, 10000
Salma Ali	Haier Dryer, hair straightener	Hussain Agahi Market Multan	Pannasonic Weekly Report	Pannasonic	03216543	4000, 6300



Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Q.No.1. B

03 Marks

Find out the functional dependencies of following cases in the table *Student-record*

- Roll No $\rightarrow$ Name
- Name $\rightarrow$ Roll No
- Dept. $\rightarrow$ Course
- Roll No $\rightarrow$ Name, Marks
- Roll No, Name $\rightarrow$ Marks
- Department, Course $\rightarrow$ Name

Student-record

Roll No.	Name	Marks	Department	Course
001	Ahmed	58	BM	C1
002	Rasheed	65	CS	C1
003	Ahmed	58	BM	C2
004	Rasheed	65	CS	C3
005	Abdul Qadir	90	EE	C3
006	Saleem-Ul-Ullah	91	IT	C2

Q.No.2

10 Marks

Internet is the wide domain for implementing the Database application. Let's consider an online bookstore that allows you to browse and buy books, such as ABC.com. The bookstore allows you to browse books in different categories, such as computing or management, or it may allow you to browse books by author name. In either case, there is a database on the organization's Web server that consists of book details, availability, shipping information, stock levels, and on-order information. Book details include book titles, ISBNs, authors, prices, sales histories, publishers, reviews, and detailed descriptions. The database allows books to be cross-referenced: for example, a book may be listed under several categories, such as computing, programming languages, bestsellers, and recommended titles. The cross-referencing also allows Amazon to give you information on other books that are typically ordered along with the title you are interested in. For financial transactions you can provide your credit card details to purchase one or more books online. ABC.com personalizes its service for customers who return to its site by keeping a record of all previous transactions, including items purchased, shipping, and credit card details. When you return to the site, you can now be greeted by name and you can be presented with a list of recommended titles based on previous purchases.

Considering above relevant database application, explain how this application maintains data transaction according to **three level architecture of database**.

Hint: You need to mention the activities at every level (at internal level display schema of database, at Conceptual level **construct ERD and Relational Model of above process**, External level contains User Interface (show maximum two modes of data incoming))

Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Q.No.3

10 Marks

Consider the following relational database schema consisting of the four relation schemas:

Passenger (pid, pname, pgender, pcity)

Agency (aid, aname, acity)

Flight (fid, fdate, time, src, dest)

Booking (pid, aid, fid, fdate)

Apply relational algebra

- Get the complete details of all flights to Karachi.
- Get the details about all flights from Lahore to Karachi.
- Find only the flight numbers for passenger with pid 123 for flights to Lahore before 06/11/2020.
- Find the passenger names for passengers who have bookings on at least one flight.
- Find the passenger names for those who do not have any bookings in any flights.

Q.No.4.A

02 Marks

Identify and discuss problem in following concurrent transaction execution?

Note: Initial of A in Database = 1000

T1	T2
R(A)	
A=A-50	
	R(A)
	A=A+100
W(A)	
	W(A)

Q.No.4.B

04 Marks

Prove that S1 is conflict equivalent to S2.

S1	
T1	T2
R(A)	
W(A)	
	R(A)
	W(A)
R(B)	
W(B)	
	R(B)
	W(B)

S2	
T1	T2
R(A)	
W(A)	
R(B)	
W(B)	
	R(A)
	W(A)
	R(B)
	W(B)

Sir Syed University of Engineering & Technology

END SEMESTER EXAMINATIONS SPRING 2022

Q.No.4.C

04 Marks

Check for conflict serializability by using precedence diagram

Q.No.4.C.1		
S1		
T1	T2	T3
R(X)		
		R(X) W(X)
W(X)		
	R(X)	

Q.No.4.C.2		
S2		
T1	T2	T3
R(X)		
	R(Z)	
R(Z)		
		R(X) R(Y) W(X)
R(Y) W(Z) W(Y)		

Q.No.5

10 Marks

Considering Employees Table as mentioned below perform the following with the help of sub query.

- Display the employee numbers, last names and salaries of all employees who earn more than the average salary and who work in a department with any employee with a *u* in their name.
- Display the department number, last name, and job ID for every employee in the Executive department.
- Display the last name, department number, and job ID of all employees whose department location ID is 1700.

Column Name	Null?	Type
EMPLOYEE ID	NOT NULL	NUMBER(6)
FIRST NAME		VARCHAR2(20)
LAST NAME	NOT NULL	VARCHAR2(25)
EMAIL	NOT NULL	VARCHAR2(20)
PHONE NUMBER		VARCHAR2(20)
HIRE DATE	NOT NULL	DATE
JOB ID	NOT NULL	VARCHAR2(10)
SALARY		NUMBER(8,2)
COMMISSION PCT		NUMBER(2,2)
MANAGER ID		NUMBER(6)
DEPARTMENT ID		NUMBER(4)